

COMPARING EPGP SURROGATES AND FINITE ELEMENTS UNDER DOF PARITY

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ABSTRACT. We present a benchmarking study comparing a boundary-constrained Ehrenpreis–Palamodov Gaussian Process (B-EPGP) surrogate with a classical Finite Element Method using Crank–Nicolson time stepping (CN-FEM) for the two-dimensional wave equation with homogeneous Dirichlet boundary conditions. The B-EPGP construction leverages exponential-polynomial bases derived from the characteristic variety to enforce the PDE and boundary conditions exactly and employs penalized least squares to estimate the coefficients. To ensure fairness across paradigms, we introduce a degrees-of-freedom (DoF) matching protocol. Under matched DoF, B-EPGP consistently attains lower space–time L_2 -error than CN-FEM, improving accuracy by roughly one order of magnitude.

1. INTRODUCTION

Background. Partial differential equations (PDEs) form the backbone of mathematical modeling in science and engineering. Notably, the majority of rational mechanics relies on PDEs to establish macroscopic models of continua such as elastic bodies and other structures, fluids, electromagnetic fields, etc. For decades, deterministic numerical methods have been the dominant tool for approximating PDE solutions with guaranteed accuracy. Nevertheless, their computational cost scales poorly with mesh resolution and simulation time. They are also only applicable to well-posed problems, necessitating the unique solvability and continuous dependence on the data in some abstract functional spaces.

In recent years, probabilistic approaches based on machine learning (ML) have emerged as powerful alternatives to classical methods. To approximate PDE solutions directly from data, frameworks like physics-informed neural networks (PINNs) [35] use deep learning models that incorporate the governing PDEs into the regression loss function and learn solutions from data; Gaussian process (GP) regression [33] is a Bayesian non-parametric approach that models distributions over functions to perform regression with uncertainty quantification; operator-based learning approaches [25, 31] are also popular nowadays. GP regression, in particular, has been extended to incorporate linear operator constraints that fit the observed data, making it possible to encode PDE structure directly into the prior [23, 26]. The Ehrenpreis–Palamodov fundamental principle establishes that solutions of constant-coefficient linear PDEs can be represented as superpositions of exponential functions, providing a theoretical foundation for operator-learning approaches [8, 32]. Härkönen et al. [14] introduced Ehrenpreis–Palamodov Gaussian process (EPGP) priors, which leverage the aforementioned fundamental principle to construct exact solution priors for linear PDEs with constant coefficients, enabling inference from noisy or partial data. Also these methods are computationally cost effective, while achieving higher accuracy than classical numerical methods.

Contribution. These methods from machine learning facilitate rapid evaluation and follow Bayesian principles to allow calibrated uncertainty quantification. Thereby, they often replace rigorous *a priori* error analysis and stability guarantees that are typical for traditional solvers with probabilistic arguments that focus on average cases and average errors. Furthermore, fair comparisons remain elusive because the two paradigms rely on fundamentally different assumptions about accuracy, complexity, and computational cost. Thus, the aim of the present paper is to compare the two paradigms quantitatively, in a fair manner. We benchmark both approaches on the classic wave equation in two dimensions subject to homogeneous Dirichlet boundary conditions.

As an archetypal numerical method, we employ the traditional piecewise linear H^1 -finite element discretization of the spatial domain combined with Crank–Nicolson time integrator, a well-established implicit scheme that balances accuracy and stability [5, 24]. On the machine learning side, we use the EPGP method, which enforces the PDE structure and the boundary conditions to be satisfied *exactly* directly in the GP prior. In this

formulation, realizations satisfy the wave equation and boundary conditions in distribution [23, 26, 2, 14, 29], yielding surrogate solutions with quantified uncertainty. For the wave equation (1), the GP prior is chosen for ensuring compatibility with the PDE constraints [2, 14, 29]. The resulting posterior mean provides a surrogate solution, while the posterior covariance quantifies uncertainty—a feature unavailable in classical FEM. Extensions of this framework have tackled boundary value problems [13], optimal control problems [3], and high-frequency or multiscale PDEs [11, 7, 18].

To ensure fairness, we introduce a benchmarking framework based on degrees of freedom (DoF) matching. For FEM, the DoF scales with the number of spatial nodes and time steps, while for EPGP it is defined via the posterior trace of the regression operator, known as influence or hat matrix in statistics. This matching allows us to compare both methods on equal footing, evaluating accuracy, stability, computational efficiency, and error-complexity tradeoffs.

Our results demonstrate that EPGP achieves higher accuracy than FEM by one order of magnitude when matched for DoF, while providing near-instantaneous evaluations/predictions once trained. Moreover, unlike classical schemes, EPGP naturally quantifies uncertainty in data-driven fashion, offering insights into model reliability. Taken together, these findings highlight the complementary strengths of deterministic discretizations and probabilistic surrogates. While FEM remains the workhorse for robust and general-purpose PDE simulation, EPGP and related methods offer new opportunities for higher accuracy, faster evaluation, and uncertainty quantification.

This work presents a systematic benchmarking study between deterministic numerical solvers and probabilistic learning-based approaches for partial differential equations (PDEs). Building on [20], we propose a boundary-constrained Ehrenpreis–Palamodov Gaussian Process (B-EPGP) formulation that enforces PDE structure exactly while allowing efficient inference from limited data. Unlike conventional numerical schemes, the proposed method offers exact PDE satisfaction, uncertainty quantification, and rapid evaluation once trained. Our contribution lies in:

- (1) constructing a deterministic B-EPGP basis that satisfies both the governing wave equation and Dirichlet boundary conditions;
- (2) developing a regularized least-squares fitting framework for stable coefficient estimation;
- (3) introducing a degree-of-freedom (DoF) matching criterion that enables fair comparison with a Finite Element Crank–Nicolson (CN-FEM) solver.

The results demonstrate that B-EPGP achieves accuracy that is one order of magnitude higher than that of CN-FEM under matched DoF, with significantly reduced computation time and built-in uncertainty quantification.

Relevance. The relevance of this work stems from the growing need for efficient and interpretable solvers for PDE-governed systems in science and engineering. Traditional numerical solvers, such as finite element and finite difference methods, offer stability and convergence guarantees but often suffer from high computational costs from fine mesh resolutions and long-time simulations. Meanwhile, emerging machine learning models, including physics-informed neural networks (PINNs) and operator learning frameworks, show promise in approximating PDE solutions directly from data but typically lack theoretical guarantees and rigorous error analysis. The Ehrenpreis–Palamodov Gaussian Process framework bridges these paradigms by embedding the analytical structure of PDEs within a probabilistic model. By comparing B-EPGP against CN-FEM under controlled conditions, our study provides insights into the trade-offs between deterministic accuracy, computational efficiency, and probabilistic uncertainty quantification—a highly relevant aspect of modern scientific computing.

Problem setup. Our work addresses this gap through a careful benchmarking study that contrasts a classical variational solver with a probabilistic surrogate grounded in PDE structure. As a canonical test case, we consider the two-dimensional linear wave equation with constant coefficients on a rectangular domain $\Omega = (0, L_1) \times (0, L_2) \subset \mathbb{R}^2$ with homogeneous Dirichlet boundary conditions:

$$(1) \quad \begin{cases} u_{tt}(x, y, t) = c^2 (u_{xx}(x, y, t) + u_{yy}(x, y, t)), & (x, y) \in \Omega, t > 0, \\ u(x, y, t) = 0, & (x, y) \in \partial\Omega, t > 0, \\ u(x, y, 0) = u_0(x, y), \quad u_t(x, y, 0) = v_0(x, y), & (x, y) \in \Omega, \end{cases}$$

where $c > 0$ is the wave speed, u_0 is the prescribed initial displacement, and v_0 is the initial velocity field.

The goal is to construct accurate surrogate and numerical approximations of $u(x, y, t)$ satisfying the above PDE and boundary conditions. The two methods are considered:

- (i) the boundary-constrained Ehrenpreis–Palamodov Gaussian Process (B-EPGP)[20], which uses exact analytical basis functions derived from the PDE’s characteristic variety;
- (ii) the Finite Element Crank–Nicolson (CN-FEM) method, a well-established variational approach based on the method of lines ensuring second-order accuracy in both space and time.

The wave equation is a fundamental model for wave propagation and an excellent benchmark for comparing numerical and learning-based solvers. In fact, the wave equation is a canonical benchmark for numerical methods because its dynamics are well understood, the solution is smooth for smooth data and accurate discretizations are known. Our goal is to use this equation as a controlled setting to compare classical numerical schemes with machine learning methods based on Gaussian processes, with the goal of assessing accuracy, stability, and efficiency under matched degrees of freedom.

Previous work. Recent years have witnessed substantial interest in integrating machine learning with PDE solvers. Physics-Informed Neural Networks (PINNs) [35], recent contributions such as Gaussian Process regression with PDE constraints [23, 14], and neural operator approaches [30] have significantly advanced the integration of machine learning techniques with the modeling of physical systems. While PINNs incorporate PDE residuals into the neural loss, their optimization procedure is computationally intensive and prone to convergence issues. Gaussian Process (GP) approaches offer uncertainty quantification and analytical structure but often require linearity assumptions. The Ehrenpreis–Palamodov theorem [8, 32] provides a foundation for constructing exact solution priors for linear PDEs with constant coefficients, enabling the development of EPGP priors [14]. Extensions to boundary value problems [20] and controlled systems [3] have recently been proposed. However, a rigorous comparison between such probabilistic PDE solvers and classical numerical schemes remains largely unexplored. Our work addresses this gap by benchmarking B-EPGP against CN-FEM using a systematic DoF-matched framework.

Outline. The remainder of this paper is organized as follows. Section 2 presents the methodological details of both the proposed B-EPGP approach and the finite element method paired with Crank–Nicolson integration we implement. Section 3 describes the numerical experiment design, including benchmark configurations and evaluation metrics. Section 4 discusses the quantitative and qualitative results, followed by Section 5, which summarizes the conclusions of our work and outlines potential future research directions.

2. METHODOLOGY

In this study, we compare two distinct approaches for solving the two-dimensional wave equation: the boundary Ehrenpreis–Palamodov Gaussian process (B-EPGP) technique [20] and the classical Finite Element Method (FEM) with Crank–Nicolson time stepping [9, 28]. Both frameworks are designed to ensure fairness through degrees-of-freedom (DoF) matching and are implemented under identical boundary and initial conditions to evaluate performance in terms of accuracy, stability, and computational cost.

The B-EPGP method [20] extends the Ehrenpreis–Palamodov theorem [32] and the EPGP algorithm [14] to construct exact solution bases for linear partial differential equations with constant coefficients that also satisfy boundary constraints. For the wave equation (1), the characteristic variety determines the set of admissible exponential-polynomial functions, which are modified to enforce homogeneous Dirichlet conditions. The resulting B-EPGP basis functions thus satisfy the governing PDE and boundary conditions *exactly/symbolically*, leading to high accuracy and stability without requiring numerical discretization of the differential operator. The coefficients of the basis expansion are obtained through a regularized least-squares regression with the usual ridge penalty [19], ensuring robustness against sampling noise and numerical instability. This framework maintains the interpretability of Gaussian processes while achieving the computational efficiency of a deterministic solver.

The classical Finite Element Method (FEM) combined with Crank–Nicolson time stepping serves as a deterministic benchmark. The spatial domain is discretized using linear triangular H^1 -elements, leading to a semi-discrete system of equations with properly assembled mass and stiffness matrices. The Crank–Nicolson scheme provides second-order accuracy for the homogeneous wave equation in both space and time on sufficiently regular solutions, i.e.,

$$(2) \quad \|u - u_{h,\Delta t}\|_{L^\infty((0,T),L^2(\Omega))} = \mathcal{O}((\Delta t)^2 + h^2) \text{ as } \Delta t, h \rightarrow 0,$$

and is unconditionally stable, i.e., no Courant-Friedrichs-Levy-type relation between the time step Δt and the space step h is required for convergence. This implicit formulation enables larger time steps without compromising stability, making FEM a robust and well-established numerical approach for time-dependent PDEs.

2.1. B-EPGP.

I think there is a lot of novelty in here in how coefficients are determined. We might advertise this novelty more. (I did not want to rewrite too much of the paper.)

@Markus: Please feel free to elaborate.

Linear partial differential equations with constant coefficients possess a rich mathematical structure that can be solved by exploiting for constructing exact solution methods. The foundation of our approach rests on the Ehrenpreis–Palamodov theorem [15, 20], which characterizes the solution space of such equations in terms of exponential-polynomial functions. Consider a linear PDE with constant coefficients of the form $A(\partial)u(x) = 0$, where $A \in \mathbb{C}[\partial]$ and $\partial = (\partial_1, \dots, \partial_n)$, is the partial differential operator and $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. The characteristic variety of A is defined by $V = \{z \in \mathbb{C}^n : A(z) = 0\}$, where $A(z)$ is obtained by substituting $\partial_j \mapsto z_j$ into the differential operator. The Ehrenpreis–Palamodov theorem establishes the solutions to linear PDEs can be represented using exponential-polynomial functions whose frequencies lie on this characteristic variety. The original EPGP approach [15] leverages the Ehrenpreis–Palamodov theorem to construct exact solutions for linear PDEs.

If the characteristic variety V has no multiplicities, solutions of $A(\partial)u(x) = 0$ can be approximated with solutions of the form $u(x) = \sum_{j=1}^r w_j e^{x \cdot z_j}$, where $z_j \in V$ are frequencies from the characteristic variety and w_j are coefficients to be determined from the data. Due to the linearity of the PDE, any linear combination of the exponential functions with frequencies from V automatically satisfy the differential equation exactly. The original EPGP approach [14] treats the coefficients w_j as unknown probabilistic quantities giving a Gaussian process formulation. To simplify our formulation, we will ignore the probabilistic approach here and only use a direct regression method that computes the mean functions of the Gaussian process. While EPGP essentially produces solutions in full space, practical applications involve boundary conditions that constrain the solution on the domain boundary $\partial\Omega$. The goal of [20] is to construct basis functions that simultaneously satisfy both the differential equation and prescribed boundary constraints. For rectangular domains and Dirichlet/Neumann boundary conditions, this leads to a natural spectral decomposition, as we explain below.

Following [20], the B-EPGP method proceeds with an algebraic calculation aiming to modify exponential solutions with frequencies from the characteristic variety to satisfy boundary constraints. By explicit computation we can see that the wave equation (1) with differential operator $A(\partial) = \partial_t^2 - c^2(\partial_x^2 + \partial_y^2)$ has the characteristic variety $V = \{(\xi_0, \xi_1, \xi_2) \in \mathbb{C}^3 : \xi_0^2 = c^2(\xi_1^2 + \xi_2^2)\}$.

Starting from the exponential solutions $e^{\xi_0 t + \xi_1 x + \xi_2 y}$ with $(\xi_0, \xi_1, \xi_2) \in V$, the B-EPGP algorithm constructs linear combinations that satisfy the given boundary conditions. For rectangular domains with Dirichlet conditions, [20] calculates the B-EPGP basis

$$\exp\left(\pm c\sqrt{-1}\sqrt{\frac{j^2}{L_1^2} + \frac{k^2}{L_2^2}}\pi t\right) \sin\left(\frac{j\pi x}{L_1}\right) \sin\left(\frac{k\pi y}{L_2}\right)$$

for $j, k \in \mathbb{Z}$. For the Neumann boundary condition, the sine functions are replaced by cosine functions.

Assuming zero initial velocity $v_0 \equiv 0$ in (1) and expressing the complex exponential in the Cartesian form using Euler's identity $\exp(\sqrt{-1}\omega t) = \cos(\omega t) + \sqrt{-1}\sin(\omega t)$, the initial condition $\frac{\partial u}{\partial t}\big|_{t=0} = 0$ eliminates the sine term, leaving only the cosine component in the temporal basis. The basis functions are

$$\varphi_{j,k}(x, y, t) = \sin\left(\frac{j\pi x}{L_1}\right) \sin\left(\frac{k\pi y}{L_2}\right) \cos(\omega_{j,k} t)$$

where $\omega_{j,k} = c\sqrt{(j\pi/L_1)^2 + (k\pi/L_2)^2}$. These $\varphi_{j,k}(x, y, t)$'s build a basis of the solution space of our PDE, including boundary conditions. We employ the ridge regression [19] approach to obtain a linear combination over a finite subset of these basis elements. This ensures that our approximate solutions satisfy both the PDE and boundary conditions in (1) *exactly*.

confused
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We seek to approximate a solution to (1) with the finite-dimensional ansatz

$$u_N(x, y, t) = \sum_{j=1}^N \sum_{k=1}^N w_{j,k} \varphi_{j,k}(x, y, t).$$

We determine the coefficients $\{w_{j,k}\}$ by fitting to the initial condition $u(x, y, 0) = u_0(x, y)$ for $(x, y) \in \Omega$:

$$u_N(x, y, 0) = \sum_{j=1}^N \sum_{k=1}^N w_{j,k} \varphi_{j,k}(x, y, 0).$$

Instead of fitting over the entire domain Ω , we discretize the problem as a standard linear regression task by randomly selecting m spatial points $S = \{(x_i, y_i)\}_{i=1}^m \subset \Omega$ following a sampling strategy described below. At each spatial grid point (x_i, y_i) , we observe

$$u_i = u_0(x_i, y_i) + \varepsilon_i, \quad i = 1, \dots, m$$

where u_i represents the observed initial displacement data at a point (x_i, y_i) and $\varepsilon_i \sim N(0, \sigma^2)$ are independent random disturbances. Evaluating the basis function at the grid points yields the linear regression model

$$\mathbf{u} = \Phi \mathbf{w} + \varepsilon$$

where $\mathbf{u} = [u_0(x_1, y_1), \dots, u_0(x_m, y_m)]^T \in \mathbb{R}^m$ contains the observed initial condition values, the vector of coefficients $\mathbf{w} = [w_{1,1}, w_{1,2}, \dots, w_{N,N}]^T \in \mathbb{R}^{N^2}$ denotes the weights of the model, $\Phi \in \mathbb{R}^{m \times N^2}$ is the design matrix with entries

$$\Phi_{i,(j,k)} = \sin\left(\frac{j\pi x}{L_1}\right) \sin\left(\frac{k\pi y}{L_2}\right),$$

and $\varepsilon \in \mathbb{R}^m$ represents the error vector.

argument missing here that if you do MLE you obtain the same as the problem below with $\lambda = \sigma^2 - M$:
See my response in the comments

Under the Gaussian noise assumption $\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, the log-likelihood of $\mathbf{w}$ given the data is

$$\log p(\mathbf{u} | \mathbf{w}) \propto -\frac{1}{2\sigma^2} \|\Phi \mathbf{w} - \mathbf{u}\|_2^2.$$

If, in addition, we impose a Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \lambda \sigma^2 \mathbf{I})$ while assuming independence from the error term, the negative log-posterior (up to an additive constant) becomes

$$\frac{1}{2\sigma^2} \|\Phi \mathbf{w} - \mathbf{u}\|_2^2 + \frac{\lambda}{2\sigma^2} \|\mathbf{w}\|_2^2.$$

Maximizing the posterior, or equivalently minimizing its negative, yields the regularized least-squares problem

$$(3) \quad \underset{\mathbf{w} \in \mathbb{R}^{N^2}}{\text{minimize}} \left\{ \frac{1}{2} \|\Phi \mathbf{w} - \mathbf{u}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \right\}.$$

Thus, the regularization parameter λ can be interpreted as the noise variance scaling in the Gaussian model. This equivalence between ridge regression and maximum *a posteriori* estimation under a Gaussian prior is well established in the statistical learning and machine learning literature [4, 16, 39]. The first term enforces fidelity to the observed data, i.e., controls the squared bias, while the second penalizes large coefficients to reduce the variance, thus promoting smoothness and numerical stability.

The objective function (3) is strictly convex, yielding the unique minimizer

$$(4) \quad \hat{\mathbf{w}} = (\Phi^T \Phi + \lambda \mathbf{I})^{-1} \Phi^T \mathbf{y}.$$

The inverse is efficiently computed using Cholesky decomposition as $\Phi^T \Phi + \lambda \mathbf{I}$ is positive definite.

To reduce sensitivity of the particular choice of grid points and improve robustness, we employ multiple samplings with coefficient averaging:

- Generate R independent uniformly distributed realizations of the grid points $S^r = \{(x_{ir}, y_{ir})_{i=1}^m\}$, $r = 1, \dots, R$ using the Latin hypercube design. **how? @Obed: explain/provide reference**
- For each set S^r , solve (3) for each configuration to obtain coefficient vectors $\{\mathbf{w}^{(r)}\}_{r=1}^R$.
- Then, we average the solution $\bar{\mathbf{w}} = \frac{1}{R} \sum_{r=1}^R \mathbf{w}^{(r)}$.

This procedure reduces sampling variance while maintaining computational efficiency. @All: How does this effect the DoF? Technically, DoF should increase by a factor of up to R , shouldn't they?

Given the fitted coefficients $\bar{\mathbf{w}}$, the solution at any time $t > 0$ and spatial location (x, y) is predicted as

$$u(x, y, t) = \sum_{j=1}^N \sum_{k=1}^N \bar{w}_{j,k} \sin\left(\frac{j\pi x}{L_1}\right) \sin\left(\frac{k\pi y}{L_2}\right) \cos(\omega_{j,k} t)$$

which can be vectorized as

$$u(x, y, t) = \Phi(x, y) \text{diag}(\cos(\omega_{j,k} t)) \bar{\mathbf{w}}$$

where $\Phi: \Omega \rightarrow \mathbb{R}^{1 \times N^2}$ contains the spatial basis evaluated at (x, y) .

2.2. Finite Element Method with Crank–Nicolson intergrator. Using the classical weak formulation of second-order hyperbolic equations [36, 9, 28], we represent the wave equation in the Hilbert space $V = H_0^1(\Omega)$ endowed with the bilinear form $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx$ for all $u, v \in V$. On the strength of Poincaré–Friedrichs inequality, the latter constitutes a norm equivalent with the standard Sobolev norm on H^1 . By letting $H = L^2(\Omega)$, we obtain the Gelfand triple (V, H, V') , and the weak formulation seeks $u \in C^2([0, T], V') \cap C^1([0, T], H) \cap C^0([0, T], V)$ such that for all $v \in V$

$$(5) \quad \langle \partial_{tt} u(t), v \rangle_H + c^2 a(u(t), v) = 0$$

for $t \in [0, T]$ with initial data $u(0) = u_0$, $u_t(0) = v_0$. This formulation implies the energy conservation identiy

$$\frac{dE(t)}{dt} = 0 \text{ or } E(t) \equiv E(0)$$

where

$$E(t) = \frac{1}{2} (\|u_t(t)\|_H^2 + c^2 \|\nabla u(t)\|_H^2)$$

is the energy of the homogeneous system [36, 10].

@Obed, Samit: Please make all vectors and matrices bold.

Let $\mathcal{T}_h$ be a conforming triangulation of Ω with the mesh size h , and let $V_h \subset V$ be the finite element space defined as

$$V_h = \{v_h \in V : v_h|_K \in \mathcal{P}_1(K) \text{ for each } K \in \mathcal{T}_h\},$$

where $\mathcal{P}_1(K)$ denotes the space of polynomials on K of up to degree 1. We approximate $u(\cdot, t)$ by

$$(6) \quad u_h(x, t) = \sum_{i=1}^N U_i(t) \varphi_i(x),$$

where $\{\varphi_i\}_{i=1}^N$ is the nodal basis of V_h . Substituting (6) into the weak form (5) and taking $v_h = \varphi_j$ yields the semi-discrete finite element system

$$(7) \quad \mathbf{M} \ddot{\mathbf{U}}(t) + c^2 \mathbf{K} \mathbf{U}(t) = 0,$$

where $\mathbf{U} = (U_1, \dots, U_N)^T \in \mathbb{R}^N$ is the vector of nodal values, $\mathbf{M}$ is the constant mass matrix with the entries

$$M_{i,j} = \int_{\Omega} \varphi_i(x, y) \varphi_j(x, y) \, dx \, dy$$

and $\mathbf{K}$ is the constant stiffness matrix with the entries

$$K_{i,j} = \int_{\Omega} \nabla \varphi_i(x, y) \cdot \nabla \varphi_j(x, y) \, dx \, dy.$$

For each triangular element T_e of area A_e , the element of the mass and stiffness matrix are defined as

$$\mathbf{M}_e = \frac{A_e}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{K}_{e,(i,j)} = \frac{1}{4A_e} (b_i b_j + c_i c_j),$$

respectively, where $b_i = y_{i+1} - y_{i-1}$ and $c_i = x_{i-1} - x_{i+1}$ (with cyclic indexing) [6, 28]. The global matrices are assembled by summing contributions from all elements.

Let $t_n = n(\Delta t)$ with the uniform stepsize $\Delta t := \frac{T}{N_t}$ for $n = 0, 1, \dots, N_t$, and denote $\mathbf{U}^n \approx \mathbf{U}(t_n)$. Applying the Crank–Nicolson scheme to the semi-discrete system (7) gives

$$\mathbf{M} \frac{\mathbf{U}^{n+1} - 2\mathbf{U}^n + \mathbf{U}^{n-1}}{(\Delta t)^2} + c^2 \mathbf{K} \frac{\mathbf{U}^{n+1} - \mathbf{U}^{n-1}}{2} = 0.$$

Introducing $\alpha = \frac{c^2(\Delta t)^2}{2}$, we can rewrite the update as

$$(\mathbf{M} + \alpha \mathbf{K}) \mathbf{U}^{n+1} = (2\mathbf{M} - \alpha \mathbf{K}) \mathbf{U}^n - \mathbf{M} \mathbf{U}^{n-1}$$

The Crank–Nicolson scheme is unconditionally stable for the homogeneous wave equation, allowing larger time steps compared to explicit methods without stability restrictions. Also, on regular solutions, namely, $u \in C^3([0, T], L^2(\Omega)) \cap C([0, T], H^2(\Omega))$ and regular meshes, this method achieves second-order accuracy in both space and time, providing an $\mathcal{O}(h^2 + (\Delta t)^2)$ in convergence in $C^0([0, T], L^2(\Omega))$. Note that since the left-hand side of the matrix $\mathbf{M} + \alpha \mathbf{K}$ remains constant throughout the implementation, it can be factorized only once at the beginning, making each time step computationally efficient.

3. NUMERICAL EXPERIMENTS

We benchmark our proposed method against the classical Crank–Nicolson finite element method (CN-FEM) on two representative test problems. Our experiments are designed to ensure a fair comparison by matching for model complexity based on degrees of freedom (DoF) generating high-accuracy reference solutions, and reporting both accuracy and efficiency metrics.

Because B-EPGP and CN-FEM have fundamentally different computational structures, we compare them under matched degrees of freedom (DoF). For B-EPGP, we define the (effective) DoF as

$$\text{DoF}_{\text{EP}} = \text{tr}(\Phi^T \Phi + \lambda \mathbf{I})$$

consistent with the formulation of effective degrees of freedom in linear regression, smoothing and regularization, where the degrees of freedom is expressed as the trace of the associated smoothing or hat matrix [17, 38, 22]. This formulation quantifies the model’s flexibility by measuring the sensitivity of the fitted values to the observed data and incorporates the influence of the regularization parameter λ . For CN-FEM with homogeneous Dirichlet boundary conditions, the DoF depends on both spatial and temporal discretization which is given by

$$\text{DoF}_{\text{CN}} = (n - 1)^2 \times N_t$$

where n denotes the number of spatial points per coordinate direction, yielding $(n - 1)^2$ interior nodes, and $N_t = T/\Delta t + 1$ represents the number of time steps. This definition follows the usual finite element interpretation of global degrees of freedom as the number of independent nodal unknowns remaining after enforcing boundary constraints [27].

To ensure a fair comparison between the two solvers, we select parameters $(n, \Delta t)$ such that $\text{DoF}_{\text{CN}} \approx \text{DoF}_{\text{EP}}$. To maintain consistency between spatial and temporal discretizations, we couple the time step to the spatial resolution by setting $\Delta t = T/n$. This choice yields $N_t = n + 1$, eliminating the explicit dependence on T in the degree of freedom relation. Substituting into the Crank–Nicolson formulation gives the cubic equation

$$(n - 1)^2(n + 1) = \text{DoF}_{\text{EP}}$$

which is solved for the real positive root to obtain n , the corresponding Δt then follows directly. Comparing computational methods at matched degrees of freedom provides a consistent basis for evaluating accuracy and cost. A similar approach has been adopted by [37], who compared the Fourier–Galerkin and finite element methods under equivalent numerical resolution. High accuracy reference solutions were computed using the CN-FEM on fine grids (400×400) with small time steps ($\Delta t \approx 0.0025$). These solutions serve as ground truth for error evaluation.

For sufficiently smooth solutions of the linear wave equation, piecewise linear FEM in space combined with the Crank–Nicolson scheme in time is second-order accurate in both space and time. The global discretization error satisfies an *a priori* bound of the form

$$\max_{t_i} \|u(\cdot, t_i) - u_h^i\|_{L^2(\Omega)} \leq C(h^2 + (\Delta t)^2),$$

where C is independent of h and Δt [28, 1, 12]. On the unit square, $h \approx 1/n$, and under our DoF-matching rule $\Delta t = 1/n$, the bounds scales as $h^2 + (\Delta t)^2 = 2/n^2$. Let n_{ref} denote the fine reference grid, while $n \ll n_{\text{ref}}$ corresponds to coarser grids used in the experiments. Then asymptotic relative errors becomes

$$\frac{h^2 + (\Delta t)^2}{h_{\text{ref}}^2 + (\Delta t_{\text{ref}})^2} = \left(\frac{n_{\text{ref}}}{n}\right)^2$$

indicating the reference solution is $(n_{\text{ref}}/n)^2$ times more accurate in the asymptotic regime. As an illustration, with $n_{\text{ref}} = 400$ and representative experimental grid $n = 28$, the ratio is $(400/28)^2 \approx 2.04 \times 10^2$ meaning the reference solution's theoretical error bound is roughly two orders of magnitude smaller than that of the coarser DoF-matched run. This justifies treating the fine-grid CN-FEM solution as a numerically exact reference in subsequent error analyses. **@All: Maybe we should do L^∞ in time and L^2 in space instead/additionally?** To ensure consistent quantitative comparison between the CN-FEM and B-EPGP, we evaluate the discrepancy between the numerical and reference solution using space-time L^2 error norm.

Let $u_{\text{ref}}(x, y, t)$ denote the high-resolution CN-FEM solution defined on the continuous space-time L^2 -norm [9, 34] as

$$(8) \quad \|u_h - u_{\text{ref}}\|_{L^2(0,T;L^2(\Omega))} = \int_0^T \int_\Omega |u_h(x, y, t) - u_{\text{ref}}(x, y, t)|^2 dx dy dt.$$

This metric naturally arises in the analysis of time-dependent problems and provides a consistent “isotropic” measure of the overall space-time accuracy. Following the standard finite element practice [9, 28], the spatial integration is approximated via Gaussian quadrature on each triangle $T_e \in \Omega_h$. Let $(x_{e,q}, y_{e,q})$ and w_q denote quadrature points and weights, and $|T_e|$ the element area. Then for any sufficiently smooth $f(x, y)$,

$$(9) \quad \int_\Omega f(x, y) dx dy \approx \sum_{e \in \mathcal{T}_h} |T_e| \sum_{q=1}^4 w_q f(x_{e,q}, y_{e,q}).$$

Note that the Gaussian quadrature integrates piecewise quadratic functions exactly. Both CN-FEM and B-EPGP solutions, as well as the reference solution, are evaluated at the same quadrature points on the CN mesh using bilinear interpolation from a finer reference grid. Temporal integration was performed using a uniform reference time discretization with first-order (midpoint) quadrature. **Should have used Gauss quadrature.** We obtain $u_{\text{ref}}(\cdot, t)$ at any arbitrary at any time t via a linear temporal interpolation. The space-time integral in (8) becomes

$$(10) \quad \|u_h - u_{\text{ref}}\|_{L^2(0,T;L^2(\Omega))}^2 = \Delta t \sum_{n=0}^{N_t-1} \|u_h(\cdot, t_n) - u_{\text{ref}}(\cdot, t_n)\|_{L^2(\Omega)}^2$$

@Obed: It's Riemann sum (1st order accuracy), no Simpson's/trapezoid rule (2nd order accuracy). Actually, to match the space procedure, we should use Gauss quadrature. To obtain a dimensionless accuracy indicator independent of the reference amplitude, we define the relative space-time L^2 error by

$$\epsilon_{\text{rel}} = \frac{\|u_h - u_{\text{ref}}\|_{L^2(0,T;L^2(\Omega))}^2}{\|u_{\text{ref}}\|_{L^2(0,T;L^2(\Omega))}^2}.$$

I don't think there is much analogy with R^2 here. Also, that squared integral isn't really energy. This metric reflects the proportion of the total reference energy misrepresented by the numerical solver [28, 21].

We consider two representative initial conditions that highlight different aspects of wave propagation. The first one is a smooth polynomial profile,

$$u(x, y, 0) = x(1-x)y(1-y),$$

which vanishes on the boundary of the domain and, therefore, naturally satisfies the Dirichlet boundary conditions. The example serves as a well-behaved test case for assessing accuracy in the smooth setting.

The second initial condition is a Gaussian pulse,

$$u(x, y, 0) = \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right)$$

with parameters $x_0 = 0.3$, $y_0 = 0.7$ and $\sigma^2 = 0.08$. **The function does not satisfy the boundary conditions. Maybe we should use Friedrichs mollifier instead? <https://en.wikipedia.org/wiki/Mollifier>** This

configuration corresponds to a localized disturbance placed asymmetrically within the domain, making it a more challenging test case compared to the smooth polynomial initial condition.

4. RESULTS

Tables 1 and 2 report the quantitative comparison between the proposed B-EPGP method and the Crank–Nicolson finite element method (CN-FEM) under matched degrees of freedom (DoF). Both solvers were tested on the unit square domain $\Omega = (0, 1)^2$, the wave speed $c = 1.0$, homogeneous Dirichlet boundary conditions, and terminal time $T = 1.0$. The B-EPGP method consistently achieves smaller space-time L^2 -errors and relative errors, while requiring less computation time.

TABLE 1. Space-time L^2 and relative errors for CN-FEM and B-EPGP under matched DoF.

Initial Condition	CN Error	B-EPGP Error	CN Relative Error	B-EPGP Relative Error	Improvement
Polynomial	3.17×10^{-3}	2.44×10^{-4}	13.13%	1.01%	$12.97\times$
Gaussian pulse	4.86×10^{-2}	7.80×10^{-3}	49.99%	8.02%	$6.23\times$

TABLE 2. Computation time comparison between CN-FEM and B-EPGP.

Initial Condition	CN Time	B-EPGP Time	Speed-up
Polynomial	0.11s	0.04s	$2.8\times$ faster
Gaussian pulse	0.08s	0.04s	$2.0\times$ faster

Both configurations were designed to operate under nearly identical numerical resolution for a fair comparison. The CN-FEM discretization used $n_x = n_y = 28$ spatial intervals with $\Delta t = 0.0357$ and 29 time levels, corresponding to approximately 21,141 degrees of freedom. The B-EPGP model employed $N = 10$ frequency modes in each spatial direction, $n = 1,000$ sample points for basis construction, and ensemble averaging over $R = 20$ independent runs, resulting in 20,259 degrees of freedom (a 4.4% DoF mismatch). @Obed: Need to argue that DoF scale linearly with R with a factor 1. I don't think it's generally true. When the design matrix of any single replication has full column rank, this would obviously be impossible. But maybe it's true if DoF of each rep is small compared to m .

Despite comparable resolution, B-EPGP achieves up to $13\times$ smaller space-time L^2 -error and runs $2\text{--}3\times$ faster. The relative error quantifies the fraction of the reference field energy that is not captured by the numerical solver. Hence, the reduction from 13% to 1% (polynomial) and from 50% to 8% (Gaussian pulse) demonstrates that B-EPGP bad wording! L^2 is not energy here. And even if it was, the energy could be the same but the error could be huge. One would need an R^2 argument instead preserves nearly all of the physical energy represented in the reference solution. The identical reference space-time norm ($\|u_{\text{ref}}\|_{L^2(0,T;L^2(\Omega))} = 2.4158 \times 10^{-2}$) confirms that both solvers were evaluated using consistent quadrature and temporal alignment.

Figures ?? and ?? illustrate the spatiotemporal evolution of the wave field for the two initial conditions (polynomial and Gaussian pulse) at snapshot times $t = 0.0, 0.25, 0.5, 0.75, 1.0$. Each row corresponds respectively to the reference solution (top), the proposed B-EPGP approximation (middle), and the CN-FEM solution (bottom).

The polynomial case exhibits a smooth standing wave pattern that is symmetric about the domain center. B-EPGP reproduces the reference contours with remarkable fidelity; the phase alignment, amplitude, and nodal structure remain virtually indistinguishable across all time frames. In contrast, CN-FEM preserves the overall mode structure but shows a gradual reduction in amplitude and a small delay in phase after $t = 0.5$. This is visible as a slight shift of the contour lines and mild smoothing near the points of maximum displacement, where the wave amplitude is largest. The smoother appearance of the CN-FEM field reflects its inherent numerical dissipation from the implicit time stepping scheme. These visual patterns are consistent

with the quantitative results where the relative space time error decreases from 13.1% for CN-FEM to 1.0% for B-EPGP, corresponding to about a thirteenfold improvement in accuracy.

The Gaussian pulse initial condition presents a more localized and dynamic disturbance that produces expanding wavefronts interacting with the domain boundaries. B-EPGP accurately reproduces the circular wavefront geometry, the gradual decay in amplitude, and the reflected patterns near the boundaries. Small oscillatory ripples appear behind the main front, a characteristic artifact of spectral truncation that resembles mild Gibbs effects. CN-FEM captures the overall propagation but displays noticeable diffusion of the wavefront, weaker peak amplitudes, and slightly slower propagation speed. By $t = 0.75$ and $t = 1.0$, the CN-FEM fronts appear broader and less defined, with the regions of maximum amplitude slightly delayed compared to the reference solution. The qualitative trends align closely with the numerical findings.

Across both test cases, the B-EPGP method demonstrates clear advantages in phase accuracy and amplitude preservation while remaining computationally efficient. Its representation effectively captures smooth oscillatory patterns, as seen in the polynomial initial condition, and retains sharper, more localized features for the Gaussian pulse. In contrast, CN-FEM remains stable and reliable but exhibits moderate numerical dissipation and slight phase delay, characteristics typical of implicit finite element schemes. Overall, the contour plots visually confirm the quantitative error analysis, showing that B-EPGP yields a more faithful reconstruction of the true wave dynamics throughout the entire simulation period.

5. CONCLUSION

The comparative analysis between B-EPGP and Crank–Nicolson FEM demonstrates the potential of operator-informed Gaussian process methods for solving partial differential equations both efficiently and accurately. Under matched degrees of freedom, B-EPGP consistently attains smaller space-time L^2 -errors and shorter computation times. For the smooth polynomial initial condition, B-EPGP reduces the space time error from 3.17×10^{-3} to 2.44×10^{-4} , corresponding to an improvement of approximately $12.97\times$ relative to FEM. For the Gaussian pulse, which represents a more localized and asymmetric disturbance, B-EPGP achieves a $6.23\times$ improvement, lowering the error from 4.86×10^{-2} to 7.80×10^{-3} . Furthermore, visually, the B-EPGP solutions capture the high-frequency features of the reference solution with remarkable clarity, whereas the FEM results tend to smooth out finer details and emphasize lower frequency components. These findings confirm that B-EPGP can reproduce the reference dynamics with near perfect fidelity while offering notable computational advantages and intrinsic uncertainty quantification.

The qualitative behavior observed in Figures ?? and ?? further supports these conclusions. For the polynomial initial condition (Figure ??), B-EPGP produces solution contours that remain virtually indistinguishable from the high resolution reference across all time steps, while FEM displays slight amplitude damping and smoothing. For the Gaussian pulse (Figure ??), B-EPGP accurately captures both the outward propagation and boundary reflections with minimal distortion, whereas FEM exhibits mild oversmoothing due to numerical “diffusion” and delayed wavefronts. We attribute the dominance of B-EPGP to its strategic choice of basis, all of whose elements are *exact* solutions to the underlying wave equation, thereby embedding the governing operator directly into the representation space.

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